Infrared Explorer Article Style Guide

Nov 10, 2026 · @Someone

How Charles Xie writes the activity articles linked from intofuture.org/ie.html, distilled so that new articles by other authors read as part of the same series.

The formula

Every article is one natural phenomenon, one real experiment the author did with a thermal camera, and one explanation of the physics behind it. The camera and the app are tools that appear in a sentence or two; the subject is always the world, never the software.

The reader is a curious teacher, student or hobbyist, not a user looking for a manual. They should finish knowing something new about tulips, metals, moths or saltwater, and feel that they could repeat the experiment themselves this afternoon.

What the articles are not: feature announcements, release notes, UI walkthroughs or lists of what the app can do. Charles's own rule, said many times: nobody cares about a feature until they see the problem it solves.

Title and tagline

The title names a question or a phenomenon, never a feature. The sidebar on ie.html shortens each one to a topic ("Thermal conductivity", "Thermogenesis of moths"), but the page itself carries the full question.

Sidebar label	Actual title on the page
Thermal conductivity	Why do metals feel colder than wood?
Thermogenesis of moths	Are moths cold-blooded?
Opening and closing of tulips	Why do tulips open in sunlight and close after sunset?
Herschel's discovery	Discovering near-infrared light with a thermal camera
Thermal gradient in saltwater	A persistent temperature gradient in saltwater

Under the title sits a one-line tagline that either teases the answer or states the surprise. The tulip article's tagline says the flower bowls collect sunlight and stay warm into the night, a "hotel" for pollinators. Write the tagline last, after you know what the experiment showed.

A quick test for a draft title: if it would still make sense with the camera removed from the story, it is the right kind of title. "Time-lapse in Infrared Explorer" fails; "How much does the sun warm a floor through a window?" passes.

Length and skeleton

Articles run from about 250 to 1,500 words; most land between 500 and 1,000. The acid-base titration piece is roughly 230 words, one figure and one open question, and it is still a complete article. Length follows the experiment, not a quota.

The order is the same every time:

1. **The phenomenon.** One or two paragraphs on what we notice in daily life and why it is puzzling.
2. **The setup.** What was used and how it was arranged, specific enough to copy.
3. **The results.** Two to four figures with a sentence or two each on what to see in them.
4. **The science.** Why it happens, taken down to molecules, heat flow or radiation.
5. **What is not settled.** A method limit, a surprise, or a question the data cannot answer yet.
6. **The closing line.** An open question or one sentence about why this matters for learning.

Longer articles use `h2` headings for stages 2 to 5. The tulip article divides by time of day instead ("During the day", "At night", "In the morning", "Conclusion"); the Herschel article uses "Reproducing Herschel's experiment", "Important notes", "Conclusion". Short articles have no headings at all.

Openings

The first paragraph never introduces the article or the app. It drops the reader into one of three hooks.

A question from daily life. "Why do metals feel colder than wood?" The tulip article opens by noting that tulips open by day and close at night, unlike most flowers, and asks why. The question is one the reader has probably had and never followed up.

A historical anecdote. The Herschel article starts with the 1800 experiment that found invisible light beyond red, then promises to redo it in a backyard. The thermal equilibration article leans on a 1985 Erickson paper on how children picture heat. History gives the experiment a lineage and tells the reader this is real science, not a toy.

An accidental discovery. "In 2010 I accidentally found…" The saltwater gradient and paper-on-cup articles begin with something Charles noticed by chance while pointing

the camera at an ordinary object. This hook works because it models the behavior the article wants from the reader: look, be surprised, investigate.

Whichever hook you use, the camera enters only at the end of the opening, as the way to get a closer look: "so I pointed a thermal camera at it."

The body

Setup is concrete enough to copy. The conductivity article says three nails, not "some metal objects". The paper-on-cup article says the camera sat three to four inches away and that hands must stay out of the frame. The Herschel article explains why there is a deliberate gap between the two paper strips (to block heat conduction and mark the visible/infrared boundary). A reader with the same camera should be able to reproduce the result from the text alone.

Results are read for the reader. Each figure gets a sentence that says what to look at and gives the number: the black center of a yellow tulip is over 2°C warmer than its petals; a red tulip shows about 5°C; breathing rate rose from 12 to 33 per minute. The figure is evidence, the sentence is the claim.

"The science" goes down a level. The explanation never stops at "metal conducts heat". It goes to molecules: entropy, hydrogen bonds between water and paper fibers, how a dark surface absorbs near-infrared while the camera sees long-wave infrared, why evaporation cools water below room temperature. One or two paragraphs, plain words, no equations unless one is the point.

Method limits are stated, not hidden. The moth article explains that the automatic color scale makes the background look blue without it actually getting colder. The tulip article adds 7 am images to rule out an emissivity artifact. The thermal equilibration article notes that water evaporates below room temperature and the simulation cannot reproduce that. The titration article admits the heat of dissolving HCl muddies the signal. These admissions are what make the rest of the article credible.

Figures and data

Every article has real data in it. None relies on a stock photo or a diagram alone.

Figure type	How it is used	Example
Visible + infrared pair	Side by side, same framing, so the reader maps colors to objects	Tulips: 13 images, each pair captioned "...seen in visible light around 2 pm" / "...seen in infrared light around 2 pm"

Temperature over time, T(t)	Shows the dynamics: warming, cooling, a rate change	Respiratory rate, thermal equilibration, cooling curves
Temperature along a line	Shows a gradient or a boundary	Herschel: "The temperature distribution in the horizontal direction"
Single thermal frame	Catches one moment when the pattern is clearest	Moth at rest vs. warming up; tulips at 11 pm
Schematic or reference chart	Only when the physics needs it	Herschel: setup drawing from The Physics Teacher, infrared spectrum chart credited to Edmund Optics

Captions are numbered and declarative: "Figure 3: Comparing the regular image and the thermal image of the warming on the dark paper." Time of day and the object are named; the camera model usually is not.

Each experiment links to a recording the reader can open and analyze. The older articles link to Telelab; new articles should link to a public recording on ie.intofuture.org with its opening view pinned (thermometers, profile lines and charts already placed). The link sits right after the figures it supports, often as its own line: "Watch an interactive video".

Screenshots for new articles come straight out of the web analyzer: the player with sensors drawn on, the T(t) chart with a time-of-day axis for time-lapse, the profile chart for gradients. Hide the UI chrome; show the frame and the chart.

Voice

Three pronouns, each with a job. "I" for a personal observation or a hunch ("I was surprised to find the tulips still warm at 11 pm"). "We" for the team and for doing the experiment together with the reader ("we can skip Herschel's slit and run it outdoors"). "You" to invite the reader in ("Imagine you were a little poor insect...", "You may be wondering...").

Numbers, not adjectives. Over 2°C, about 5°C, 30°C, 12 to 33 breaths per minute, a $200 camera, tens of thousands of "thermometers" versus Herschel's three. A sentence with a measurement beats a sentence with "significantly".

Confident about the data, honest about the explanation. Phrases like "requires some further thinking" and "we just do not have a clear picture" appear right next to firm statements of what was measured. Speculation is labeled as speculation: the pollinator hotel is offered as a hypothesis, not a finding.

Conversational, with a little humor. Short sentences, everyday words, the occasional aside. It reads like a scientist explaining something at the kitchen table, not a paper.

What he never does. No "In this article we will…". No bullet lists of features. No marketing adjectives (powerful, seamless, intuitive). No hedging every sentence. No ending with a summary of what was just said.

Where the app appears

The app gets one or two sentences per article, always as the instrument: "recorded with Infrared Explorer and a FLIR ONE Pro", "the horizontal temperature profile was taken in the app". It is named the way a biologist names a microscope.

Features are shown, not described. A time-lapse article never says "the app supports time-lapse"; it says "we recorded the floor at one frame per minute for six hours" and lets the T(t) chart with a time-of-day axis do the rest. A thermoruler article shows the profile chart of a real object and calls the tool by name once.

This is the direct answer to Charles's standing complaint: users care about applications, not UI, and a feature without a demonstrated application reads as worthless. The article is the application. If a feature cannot carry an article in this format, that is a signal about the feature, not the format.

The one exception is the pair of reflective pieces under "Articles" rather than "Activities" ("From sensing to imaging", "Post-processing images"). Those are about the instrument itself, written in the same voice, and still built on real images rather than screenshots of menus.

Closings

The last paragraph is short and does one of three things. It never recaps.

- **Leaves a question open.** Why do tulips stay closed on cloudy days? What is the gradient in saltwater doing after a week? The titration article ends with the unresolved HCl heat problem.
- **Names a limit of the method.** The camera sees long-wave infrared, so Herschel's trick cannot find ultraviolet; Ritter needed photographic paper for that.
- **Says why this matters for learning.** One sentence, often a rhetorical question: why not give these cameras to children? Why not equip every school with this power? The Herschel piece closes with tens of thousands of thermometers against Herschel's three.

The Telelab-era articles add a one-line NSF acknowledgement. Every article ends with the back link "← Back to Infrared Explorer" and nothing after it.

The corpus at a glance

The corpus at a glance

Thirteen activity articles are linked from ie.html, plus two reflective pieces about the instrument. Pages are static HTML on intofuture.org; the file name is the link target.

Topic (sidebar)	Page	What it is about
Thermal equilibration	ie-thermal-equilibrium.html	Objects drifting to room temperature; opens on Erickson's 1985 work on how children picture heat; notes that water evaporates below room temperature, which the simulation cannot reproduce
Rubber band experiment	ie-rubber-band.html	A stretched rubber band warms and a released one cools; the explanation is entropy
Thermal conductivity	ie-conduction.html	"Why do metals feel colder than wood?"; nails in different materials show conduction from a hand
Herschel's discovery	telelab-herschel.html	Redoing the 1800 experiment outdoors with a dark paper strip; a horizontal profile shows warming beyond red and nothing beyond violet
Opening and closing of tulips	ie-tulips.html	Tulip bowls stay warmer than their petals by day and warmer than the air at night; a pollinator-shelter hypothesis
Thermogenesis of moths	ie-moth.html	"Are moths cold-blooded?"; a moth warms itself before flight; a caveat about the automatic color scale
Paper on cup	ie-paper-on-cup.html	The yin-yang logo: a paper cutout on a cooled cup of water, found by chance
Hydration of paper	ie-hydration.html	Paper warms when wetted; hydrogen bonds between water and cellulose
Respiratory rate	ie-breath.html	Breathing read from the temperature at the nose; rate rose from 12 to 33 per minute after exercise
Acid-base titration	ie-titration.html	Heat of neutralization seen in the camera; about 230 words and one figure; the heat of dissolving HCl is left as an open problem
Latent heat	telelab-latent-heat.html	A temperature plateau during a phase

		change
Boiling point	telelab-boiling-point.html	Water holding at its boiling temperature while it boils
Thermal gradient in saltwater	ie-temperature-gradient.html	A vertical temperature difference in a glass of saltwater that does not go away; found by accident

The two reflective pieces are "From sensing to imaging" (ie-from-sensing-to-imaging.html) and "Post-processing images" (ie-filters.html). Both are about the instrument, written in the same voice over real images.

These one-line descriptions come from a read of the pages on 2026-11-09. Re-read a page before quoting a number from it.

Checklist before sending a draft

- [ ] The experiment was actually done, and the recording is public on ie.intofuture.org with its opening view pinned
- [ ] The title is a question or a phenomenon; it would still make sense if the camera were not in the story
- [ ] The tagline teases the result in one line
- [ ] The first paragraph is a hook (question, history or accident), not an introduction
- [ ] The setup can be copied from the text alone: materials, distances, timing, what to keep out of frame
- [ ] Two to four real figures, numbered, each with a declarative caption and a sentence saying what to see
- [ ] At least one measurement with a unit in the text, in degrees C
- [ ] "The science" goes one level deeper than the obvious explanation
- [ ] At least one limit of the method or open question is stated
- [ ] The app is named once or twice as the instrument; no feature list, no UI screenshots
- [ ] The closing is an open question, a limit or one line on learning; no recap
- [ ] 500 to 1,000 words unless the experiment needs more or less
- [ ] Byline, back link, and (if the site convention holds) the NSF acknowledgement

Page template

The articles are hand-written static HTML using the site's "v2" layout. The source is not in our repositories; Charles holds it. A draft handed over in this shape can be dropped into

the site without rework.

```
<header class="v2-hero" data-wordmark="Infrared Explorer">
  <span class="v2-eyebrow">Infrared Explorer</span>
  <h1>Why do tulips open in sunlight and close after sunset?</h1>
  <p class="v2-tagline">One line that teases the result.</p>
</header>

<section class="v2-section is-wide">
  <div class="v2-grid">
    <div class="v2-col-main">
      <a class="v2-backlink" href="ie.html">&larr; Back to Infrared Explorer</a>
      <p class="v2-byline">By Charles Xie &middot; charles@intofuture.org</p>
      <div class="v2-article-audio"><!-- podcast player, optional --></div>

      <p>Opening paragraph: the hook.</p>
      <p class="v2-pullquote">Optional one-sentence pull quote.</p>

      <h2>Section heading</h2>
      <p>Body text.</p>

      <div class="v2-screenshot-row">
        <img src="images/ie-xxx-visible.jpg" alt="...">
        <img src="images/ie-xxx-infrared.jpg" alt="...">
      </div>
      <p class="muted" style="text-align:center">Figure 1: What the pair shows,
when it was taken.</p>

      <div class="v2-model-link">
        <a href="https://ie.intofuture.org/experiment/..." target="_blank"
rel="noopener">Open the recording and analyze it yourself</a>
      </div>

      <h2>The science</h2>
      <p>...</p>

      <p>Closing paragraph.</p>
      <a class="v2-backlink" href="ie.html">&larr; Back to Infrared Explorer</a>
    </div>
  </div>
</section>
```

Conventions seen across the pages: external links carry `target="_blank" rel="noopener"`; figure captions are centered muted paragraphs starting with "Figure N:"; image pairs share one `v2-screenshot-row`; the recording link gets its own `v2-model-link` box right after the figures it supports. The exact class names for the caption's muted style should be checked against a live page before publishing, since only the

muted style should be checked against a live page before publishing, since only the structure above was confirmed.

Worked example: an experiment designed from a photo of the table

This is the article to write first. Its subject is the whole loop: photograph what is on the bench, ask an AI to design an experiment from it, do that experiment by hand, record it, and ask the AI to analyze the result. Charles asked for exactly this piece so that potential collaborators can picture what they are being invited to work on. The readers are chemistry instructors, so the article must leave them able to see a student doing the same thing.

The experiment itself is not chosen in advance. Whatever the AI proposes from the real photo is what gets done. That is the point of the article, and it is also its risk: if the first photo yields nothing worth doing, that becomes part of the story, not something to hide.

Title: Can an AI design an experiment from a photo of what is on the table?

Tagline: We photographed a lab bench, asked an AI what to do with it, did what it said, and then asked it what we had found.

1. **Hook (a question).** Every lab manual starts with a procedure. What if it started with a photo instead? One paragraph on why designing an experiment is the part students rarely get to practice.
2. **The photo.** Figure 1 is the actual photo, taken with a phone, of real materials as a student would have them: glassware, a few chemicals, a thermometer, a thermal camera, whatever is there. The caption lists the items. No staging beyond what is already on the table.
3. **The ask.** The prompt is quoted word for word. The AI's answer is shown as a figure (a screenshot of the conversation) or quoted in full: two or three proposed experiments, each with its question, steps, what to measure and a safety note. Keep the proposal that was rejected and say why: unsafe, no measurable signal, needs something not in the photo, too trivial.
4. **Choosing.** One short paragraph on why this proposal: it has a signal the thermal camera can see, it uses only what is in the photo, it fits in half an hour, it is safe to do at a bench.
5. **Doing it by hand.** The setup paragraph of any article in the series: quantities, distances, timing, what to keep out of frame. Recorded with the thermal camera; a regular thermometer or a scale beside it where that helps. Figures 2 and 3: a thermal frame or a visible-and-infrared pair, and the temperature-over-time chart with the key moments marked.
6. **Asking the AI to analyze.** What it was given (the recording, or the chart and frames)

6. **Asking the AI to analyze.** What it was given (the recording, or the chart and frames), the prompt quoted, its analysis quoted. Figure 4: a screenshot of the analysis. Then three plain sentences: what it got right, what it got wrong, what it hedged on. If it placed its own probes or ran its own calculation, show that too.
7. **The science.** The author's own explanation of the phenomenon, one level deeper than the AI went. This is where the article shows that the human still knows more than the assistant, and why that matters for teaching.
8. **What is not settled.** Where the loop broke: a proposal that ignored a material in the photo, a safety point the AI missed, an analysis that forgot there was no control. Also the real open question: do students design better experiments when an AI drafts the first one, or worse?
9. **Closing line.** The experiment came from the table, not from a textbook. Then the link to the public recording and, if possible, to the conversation itself, so the reader can repeat the loop with their own photo.

Rules specific to this article. Prompts and answers are quoted verbatim, never paraphrased in the AI's favor. The author's own hands do the experiment; the AI never touches the bench. Every figure is real: the real photo, the real conversation, the real data. The assistant is named once, as the tool, and only by a name that has been agreed; if the product name is still open, "an AI assistant we are building" is enough. Errors stay in the article: a reader who sees the AI get something wrong, and the author catch it, trusts the rest.

If the loop is repeated with a second photo or a second AI model, that is a second article, not a longer first one.